# CE205 Database and Information Retrieval

## Class 2

**Objective:** The main objective is to remove the redundant data from the base relation using normalisation process and convert the relation to desired normal form.

**Problem 1:**

Consider the relation REFRIG(MODEL#, YEAR, PRICE, MANUF_PLANT, COLOR), which is abbreviated as REFRIG(M, Y, P, N, C), and the following set of functional dependencies:

M→N
MY→P
N→C

Compute closures of M, MY, and N. Then, consider the decomposition of REFRIG into sub-relations R1(M, Y, P) and R2(M, N, C). Is this decomposition lossless? Justify your answer using the conditions of lossless decomposition.

**Solution guideline for problem 1:**

Find the following:

$M^+$= MNC

$\{MY\}^+$= MYNCP

$N^+$= NC

Justify, lossless or not? *(for conditions see slide 12 of Lec 2b)*
We have

REFRIG(M, Y, P, N, C),
R1(M, Y, P) and R2(M, N, C)

**Condition 1:** Union of attributes of R1 and R2 is {M, Y, P, N, C} which is equal to attributes of main relation {M, Y, P, N, C}.

**Condition 2:** M is common attribute between R1 and R2.

**Condition 3:** M is candidate key of R2 because M can determine all attributes of R2.

Therefore, the decomposition is lossless.

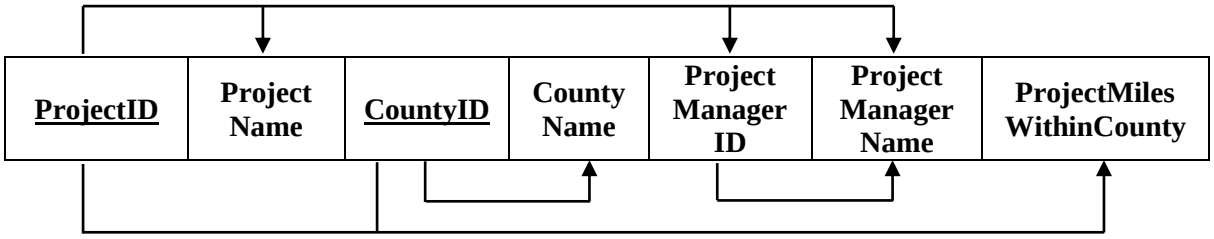
**Problem: 2(a)**

Consider the relation HIGHWAY with sample data (*the underlined attributes represent the primary key, and assume that there is no space in the attributes' name*).

| **ProjectID** | **Project Name** | **CountyID** | **County Name** | **Project Manager ID** | **Project Manager Name** | **ProjectMiles WithinCounty** |
|---|---|---|---|---|---|---|
| 1 | Road X | 1 | Wilson | M1 | Bob | 10.00 |
| 1 | Road X | 2 | Ottawa | M1 | Bob | 17.00 |
| 1 | Road X | 3 | Davis | M1 | Bob | 12.00 |
| 2 | Road Y | 3 | Davis | M2 | Sue | 23.00 |
| 3 | Bridge A | 1 | Wilson | M3 | Lee | 0.50 |
| 3 | Bridge A | 2 | Ottawa | M3 | Lee | 0.30 |
| 4 | Tunnel Q | 2 | Ottawa | M1 | Bob | 2.00 |
| 5 | Road W | 4 | Pony | M4 | Rob | 23.00 |

- *The HIGHWAY Project Table captures the data about projects and their length (in miles).*
- *Each project has a unique Project ID and Project Name.*
- *Each county has a unique County ID and County Name.*
- *Each project manager has a unique Project Manager ID and Project Manager Name.*
- *Each project has one project manager.*
- *A project manager can manage several projects.*
- *A project can span across several counties.*
- *This table records the length of the project in a county in the ProjectMilesWithinCounty column*

Based on the above assumptions, the following functional dependencies exist in the relation




- Depict full key functional dependencies, partial functional dependencies (if any), and transitive functional dependencies (if any) in the elation HIGHWAY? Provide reasons why these functional dependencies are full key, partial, and transitive?

- Redesign the relation HIGHWAY such that it must satisfy 2NF and 3NF? First, show the process of normalizing the relation in 2NF. Then, show the process of normalizing the relation in 3NF. Clearly, show all functional dependencies and keys in the resulting sub-relations.

**Solution guideline for problem 2(a):**

Let's represent the attributes by A, B, C, D, E, F, and G respectively. Now, we can write functional dependencies as

A → B E F
C → D
AC→ G
E→F

Although the primary key of the relation is given (the underlined attributes), find it using the closure of attributes.

(ABCDEFG)+ = ABCDEFG
(ACDG)+= ACDGBEF
(ACG)+= ABEFCDG
(AC)+= ABEFCDG

So, the primary key of the relation is AC.

Now answer the following questions.

i. Write the **primary key** of the relation: {AC}

ii. Write **prime attributes**: {A, C}

iii. Write **non-prime attributes**: {B, D, E, F, G}

iv. Write **proper subsets of the primary key**: {A} and {C}

v. Have you identified the **partial FD(s)** in the relation? Why are they partial FD(s)? Give reason.

A → B E F
C→ D

{A} and {C} are proper subsets of the primary key and determine non-prime attributes.

vi. Have you identified the **transitive FD(s)** in the relation? Why are they transitive FD(s)? Give reason.

E → F

The non-prime attribute is determining a non-prime attribute. Therefore, the FD is transitive.

vii. Have you identified the **full key FD(s)** in the relation? Why are they full key FD(s)? Give reason.

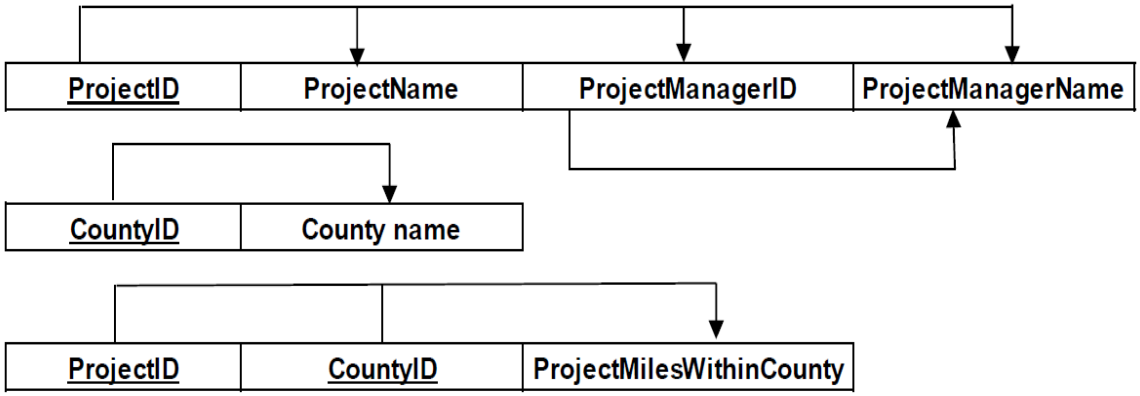
AC→ G
Only the full primary key can determine G. Therefore, it is a full key FD.

viii. Decompose the base relation to remove partial FDs **(2NF):**

To remove partial FDs, R( ABCDEFG) will be decomposed into three new relations

R1 (<u>A</u>BEF)
R2(<u>C</u>D)
R3(<u>*AC*</u>G)

We can represent them as




ix. Write the **primary and foreign key(s)** in the decomposed sub-relations(s):

A is a primary key in R1.
C is a primary key in R2.
A and C are primary and foreign keys in R3.

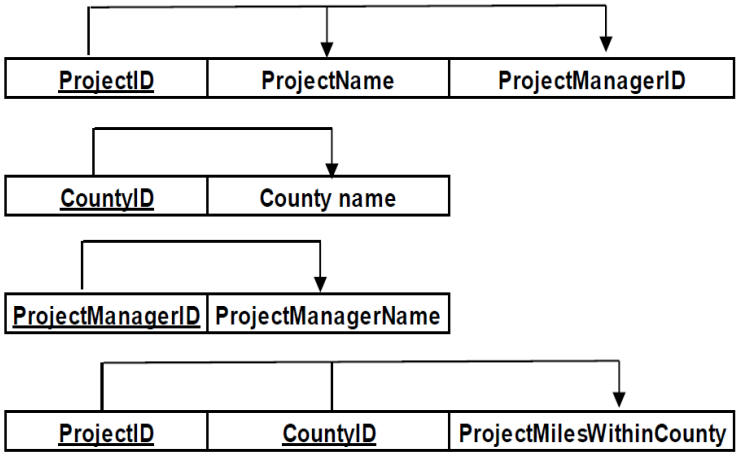
x. Decompose the sub-relation(s) to remove transitive FDs **(3NF):**

To remove transitive FD R1 will be decomposed into two new relations R11 and R12.
All other sun-relations (R2 and R3) are already in 3NF as they do not contain any transitive FDs.

So we will get

R11(AB*E* ) R12 (EF)
R2(CD)
R3(*AC*G)

We can represent them as

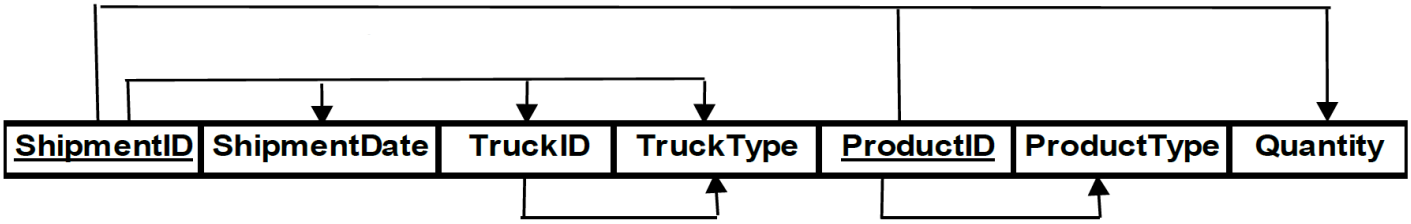



xi. Write the **primary and foreign key(s)** of the decomposed sub-relations(s):

A is a primary key in R11.
E is a foreign key in R11.
E is a primary key in R12.
C is a primary key in R2.
A and C are primary and foreign keys in R3.

**Problem 2(b):**

Repeat the same steps as mentioned in problem 2(a) for the relation given below:




**Solution:** *(follow the guideline of problem 2a)*